Xu Zheng

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SUMMARY STATEMENT

Applied Scientist candidate specializing in **LLM Agents, Time-Series Learning, and Graph-based Modeling**. Experienced in designing novel methods, conducting empirical research, etc. Skilled in **Explainable AI** and **Machine Learning**, committed to developing reliable machine learning systems that advance real-world applications.

EDUCATION

Florida International University **Miami, FL**
Ph.D. in Computer Science *01/2023 - 08/2027(Expected)*

University of Electronic Science and Technology of China **Chengdu, China**
M.E. in Control Science and Engineering *09/2018 - 06/2021*

Chongqing University **Chongqing, China**
B.E. in Electronic Science and Technology *09/2014 - 06/2018*

EXPERIENCE

DSSS, NEC Laboratories America, Inc **Princeton, NJ**
Research Scientist Intern *05/2024 - 08/2024*

- Developed an anomaly detection system combining time-series and textual data, cutting false alarms by 50% through improved cross-modal interactions.
- Designed a multi-modal fusion strategy integrating three complementary detection methods, reducing false alarms by 90% on a real-world dataset via majority voting.

Department of Image Algorithm, ZTE **Chengdu, China**
Machine Learning Engineer *07/2021 - 01/2023*

- Built an automated data collection pipeline, capturing 2,000+ images for training, and designed deep neural networks for image restoration in Under-Display Camera (UDC) systems.
- Applied pruning and quantization to optimize Raw Image Denoising Networks, improving inference speed by 25%.
- Deployed and validated optimized neural networks on Android devices using TensorFlow Lite and Snapdragon QNN frameworks.

PROJECTS

Graph for LLMs & Agents *01/2025 - Present*

- Proposed LM^{2}otifs, a graph-motif-based framework for machine-generated text detection, improving accuracy by 4% while providing interpretable multi-level explanations.
- Introduced Environment Copilot, a graph-based diagnostic framework for LLM agents that identifies error-prone action patterns, improving long-horizon task success rates by 14.7%.
- Developed a graph-based memory framework for LLM agents, achieving 15.4% and 5.47% improvements in success and progress rates over stateless baselines.

SELECTED PAPERS

- **Xu Zheng**, et al., Dongsheng Luo, "GMAIS: Graph-based Memory for Agent Inference Scaling", Under Review, 2026.
- **Xu Zheng**, et al., Dongsheng Luo, "Environment Copilot: A Graph-based Framework for Action Error Diagnosis and Decision", Under Review, 2026.
- **Xu Zheng**, et al., Dongsheng Luo, "LM^{2}OTIFS: An Explainable Framework for Machine-Generated Texts Detection", arXiv Preprint, 2025.
- **Xu Zheng**, et al., Dongsheng Luo, "Uncovering Insights of Compound Flooding with Data-Driven AI", **KDD[Acceptance Rate: 30.8%]**, 2026.
- **Xu Zheng**, et al., Dongsheng Luo, "F-Fidelity: A Robust Framework for Faithfulness Evaluation of Explainable AI", **ICLR[Acceptance Rate: 31.73%]**, 2025.
- **Xu Zheng**, et al., Dongsheng Luo, "Parametric Augmentation for Time Series Contrastive Learning", **ICLR[Acceptance Rate: 30.94%]**, 2024.
- **Xu Zheng**[*], Farhad Shirani[*], et al., Dongsheng Luo, "Towards Robust Fidelity for Evaluating Explainability of Graph Neural Networks", **ICLR[Acceptance Rate: 30.94%]**, 2024.
- **Xu Zheng**, Yali Zheng, Shubing Yang, "Generating Multiple Hypotheses for 3D Human Mesh and Pose using Conditional Generative Adversarial Nets", **ACCV**, 2022.